

WasteNet: A Structured-Pruned and EMA-Augmented RepVGG Framework for Real-Time Trash Classification

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Abstract—Achieving efficient classification of garbage boosts recycling and helps with pollution control as well as sustainable waste management. To address this challenge, we developed RepVGG-MEM, which is a lightweight and highly accurate Deep Learning model for real-time waste classification. Built on RepVGG, the model applies EMA, Mixup data augmented training, and structured pruning to optimization to maintain a strong speed performance ratio. The model and its described metrics, performance of 95.06% accuracy and a 94.5% F1 score, even with closely resembling waste categories surpass several previous versions. Moreover, its accuracy makes it ideal for edge devices and smart mobile waste management systems. RepVGG-MEM shows better performance and generalization than earlier versions. Its lightweight design makes it easy to deploy on edge devices such as smart bins and mobile platforms. This capability positions it as a promising solution for automated waste management in real-world situations.

I. INTRODUCTION

Classifying waste is important for maintaining a sustainable environment in a rapidly growing urban city and managing the increasing amounts of municipal solid waste. Improper segregation of waste means that recyclable items and hazardous materials are carelessly discarded in landfills. This, in turn, causes pollution, green house gas emission, and depletion of resources. On the other hand, proper segregation ensures materials such as plastics, organic compounds, glasses, metals, and papers are separated. This helps conservation efforts and reduces environmental harm.

The MSW in particular poses a very distinct challenge. The rapid growth of urban populations is not just due to the number of people, but also to the limited availability of public services. health infrastructure, and sustainability are lacking, as well as the sharp increase in the amount of waste. Manual sorting systems, where humans are the primary operators, are both hazardous and time-consuming. With the increase of global waste, there needs to be intelligent automation capable of

effectively dealing with waste without posing a risk. Proper waste classification not only has environmental benefits, but also supports public health and improves overall operational efficiency.

Manual sorting consumes a great deal of both time and effort, which poses a health risk to workers. Automated classification systems accurately sort waste for recycling or treatment, decrease contamination, and enable new technologies like smart bins and automated waste facilities. In short, waste classification promotes a cleaner, safer, and more sustainable urban future.

Convolutional Neural Networks (CNNs) are now the leading architecture in Deep Learning for image classification. They automatically learn spatial hierarchies of features from raw data. CNN-based VGG networks have been shown to be effective for pest identification [1]. They highlighted how deep feature extraction helps recognize subtle visual differences in agricultural images. In another study, CNNs were used for recognizing traffic signs automatically. [2]. They noted that the model performs well in real-time situations with different lighting and background conditions. In healthcare, CNN architectures have been applied to detecting brain tumors. [3]. Their work showed that CNNs capture intricate patterns in medical images with high accuracy.

These studies confirm that CNNs are highly flexible across various areas. They can consistently deliver reliable performance in image classification tasks, making them a good option for smart waste sorting systems. We suggest RepVGG-MEM for current challenges, a lightweight The Deep Learning model for real-time waste classification RepVGG backbone, this model applies Exponential Moving Average (EMA) for stable learning, Mixup generalization, and structured pruning for high accuracy, low complexity. Its smart bins, edge device, compact design, and portable real-time waste sorting in diverse environments.

II. LITERATURE STUDY

There has been new focus on classification of waste using lightweight neural networks with embedded attention due to the accuracy the models achieve using far less computing resources. Sayed et al. [4] presented the RepVGG-MEM model which is a lightweight convolutional neural network with a scale channel attention (MSCA) module. This model improves feature extraction while maintaining a compact structure with 93.26% accuracy in classification of the different categories of waste.

Hossen et al. [5] published an RWC-Net model which is dense with a Deep Learning hybrid model based on DenseNet201 and MobileNet V2. The model was trained using 5-fold cross-validation together with augmentation techniques and achieved an overall accuracy of 95.01%. The study also employed visualization techniques like Score CAM to model behavior analysis.

Tran et al. [6] proposed a combination of DenseNet161, ResNet152, and MobileNetV3 into a single model with integrated deep learning and optimized using the HHO algorithm, and created an ensemble approach. This approach combined architectural fusion with metaheuristic tuning achieving a classification accuracy of 97.45%.

Lapo and Cumbicus-Pineda et al. [7] have created a system for the detection of garbage based on the RetinaNet object detection model with a ResNet-50 backbone. The model achieved 97% accuracy after being trained on datasets containing recyclable waste. Dipo et al. [8] analyzed real-time waste classification using the YOLOv12 model. They captured images using mobile phone cameras and retrieved some from online sources. After training for 100 epochs, the model achieved a mean Average Precision (mAP) score of 78%. This was effective for implementation in mobile-based smart waste systems.

Sayed et al. [9] researched a model with InceptionV3 as its backbone and used the Multi-objective Beluga Whale Optimization (MBWO) algorithm for its refinement. The model's classification accuracy of 97.75% indicated its efficacy in performing classification tasks seamlessly.

Chhabra et al. [10] introduced an advanced convolutional neural network incorporating LeakyReLU activation and Dropout layers. With these modifications, the model was able to achieve an accuracy of 93.28% showing that the changes made were appropriate and advantageous to classification performance.

III. PROPOSED METHODOLOGY

Description of Dataset

The model was trained using three public datasets: TrashNet, Kaggle Garbage Classification, and PublicGarbageNet. This helps create a waste sorting system that works well in real-life situations. Together, these datasets have over 15,000 labeled images showing different types of waste like cardboard, electronics, plastic, paper, metal, glass, and food.

The dataset used in this study is designed to support robust and accurate waste classification across multiple cat-

egories of garbage, including plastics, metals, glass, paper, and biodegradable materials. It consists of thousands of high-resolution labeled images collected from diverse real-world environments, ensuring variability in lighting, backgrounds, and object orientation. The design of this dataset is influenced by proven approaches in other domains, such as sarcasm detection from textual data [11], lung cancer recognition using deep learning ensembles [12], and medical imaging-based fetal classification [13]. The inspiration drawn from these studies emphasizes the importance of domain-specific preprocessing, label consistency, and explainable deep learning methods for reliable classification performance. By incorporating similar standards, the dataset in our work aims to enhance the generalization capabilities of lightweight neural networks such as RepVGG-MEM for real-time deployment.

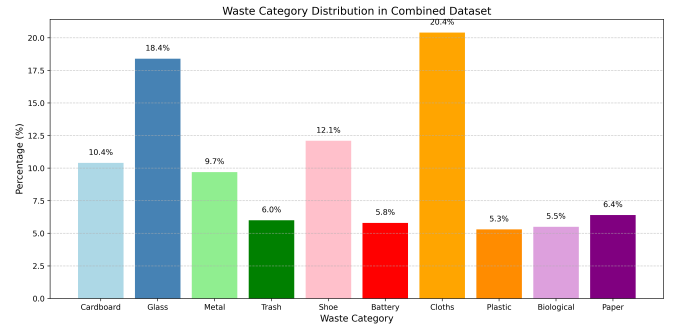


Fig. 1: Waste category distribution in the combined dataset.

The figure 1 shows the percentage distribution of different waste categories in the combined dataset. The largest share goes to Cloths at 20.4%, followed by Glass at 18.4% and Shoes at 12.1%. In contrast, categories like Plastic, Biological, and Battery make up the smallest shares, each under 6%. This distribution reveals dataset imbalance, highlighting the most and least represented categories.

The dataset was divided into three parts: 60% for training, 20% for checking how well the model is learning, and 20% for testing [14] to make sure the results are reliable.

Methods of Preprocessing

Before the classification model is trained, the input waste images go through a number of meticulously planned preprocessing steps to standardize, improve, and augment them. Increasing dataset diversity, normalizing spatial scales, and lowering noise are the goals of these operations.

TABLE I: Preprocessing Techniques

Step	Technique	Purpose
1	Resize + Crop	Standardize input size (224x224 px)
2	Gaussian Blur	Reduce noise, preserve edges
3	Grayscale Conversion	Simplify and reduce computation
4	CLAHE	Enhance contrast locally
5	Min-Max Normalization	Scale pixel values uniformly (0–255)
6	RGB Restoration	Convert back to 3-channel input
7	Horizontal Flip	Augment dataset for variation
8	Random Rotation	Improve generalization ($\pm 15^\circ$ rotation)

Gathering and Preparing Data: Several open-source waste image datasets, such as TrashNet, the Household Garbage Dataset, and image repositories hosted on Kaggle, were combined to create the dataset used in this investigation.

1. Image Resizing and Cropping: After center-cropping each input image to produce a square aspect ratio, it is resized to 224 by 224 pixels:

$$I_{\text{resized}} = \text{Resize}(\text{Crop}(I), 224 \times 224) \quad (1)$$

This preserves the central area where waste objects usually appear and guarantees consistent input dimensions throughout the network.

2. Blurring with Gaussian: A Gaussian filter is used to reduce high-frequency noise and artifacts:

$$I_{\text{blurred}} = G_{\sigma}(I_{\text{resized}}), \quad G_{\sigma} = \text{Gaussian kernel of size } (5, 5) \quad (2)$$

In order to facilitate stable feature extraction, this step smoothes the image while maintaining structural edges.

3. Conversion to Grayscale: To make further improvement easier, the blurry image is converted to grayscale:
Gray = ConvertToGrayscale(blurred)

$$I_{\text{gray}} = \text{ConvertToGrayscale}(I_{\text{blurred}}) \quad (3)$$

4. CLAHE, or Contrast Limited Adaptive Histogram Equalization: In low light, CLAHE is used to improve visibility and local contrast:

$$I_{\text{clahe}} = \text{CLAHE}(I_{\text{gray}}, \text{clipLimit} = 2.0, \text{tileGridSize} = (8, 8)) \quad (4)$$

To avoid noise over-amplification, it separates the image into tiny blocks and applies histogram equalization within each.

5. Normalization by Min-Max: The 0–255 range is used to normalize the improved grayscale image:

$$I_{\text{norm}} = \frac{I_{\text{clahe}} - \min(I_{\text{clahe}})}{\max(I_{\text{clahe}}) - \min(I_{\text{clahe}})} \times 255 \quad (5)$$

Consistent pixel value ranges across samples are guaranteed by this normalization.

6. Restoration of RGB: The normalized grayscale image is converted back to RGB since the model requires 3-channel inputs:

$$I_{\text{final}} = \text{GrayscaleToRGB}(I_{\text{norm}}) \quad (6)$$

7. Augmenting Data: To increase the size of the training dataset and lessen overfitting, several data augmentation strategies are employed:

- *Flipping Horizontally:*

$$I_{\text{flip}} = \text{Flip}(I_{\text{final}}, \text{axis} = \text{horizontal}) \quad (7)$$

- *Rotation at Random:* The image is rotated by $\pm 15^\circ$ using an affine transform:

$$I_{\text{rot}} = \text{Rotate}(I_{\text{final}}, \theta = \pm 15^\circ) \quad (8)$$

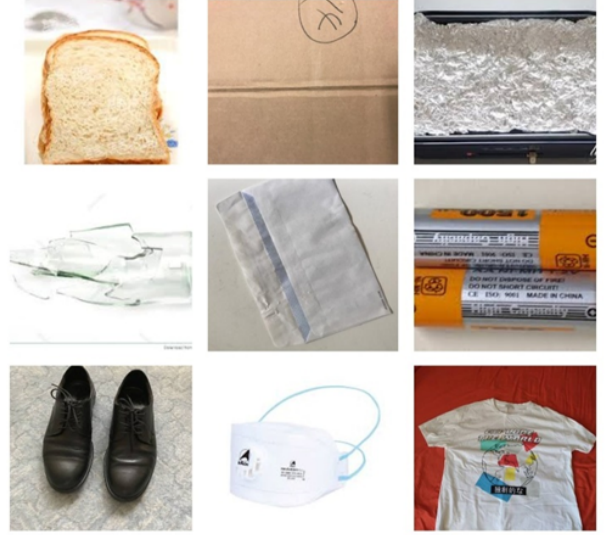


Fig. 2: Before Preprocessing



Fig. 3: After Preprocessing

General Process Flow

The RepVGG-based automated garbage classification system's high-level pipeline is depicted in Figure 2 using five consecutive blocks. Information gathering is the first step in the process, during which labeled waste images are gathered from datasets such as TrashNet and TrashBox. High-level features are then extracted from raw images using the CNN Backbone (RepVGG).

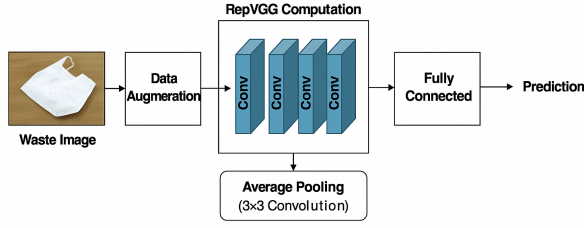


Fig. 4: Block Diagram of the Proposed RepVGG Architecture

The figure 4 shows the whole process of sorting waste using the RepVGG model. To make the model work better, the original waste image is changed in different ways like flipping, rotating, or resizing. This is called data augmentation. After the image goes through a convolution step, average pooling with a 3×3 kernel is used to make the feature maps smaller while keeping the important details. Finally, the system gives the predicted waste category after sending the output to a fully connected layer that acts as a classifier.

Waste images. Then, to improve diversity and avoid overfitting, data augmentation methods like scaling and rotation are employed. Convolutional layers are then used by the RepVGG Backbone to extract spatial features from the augmented images. The Action Module optimizes hyperparameters and feature importance using a metaheuristic algorithm (such as the Genetic Algorithm or PSO) after receiving intermediate features. By highlighting pertinent channels and suppressing less helpful ones, the Channel Attention (CA) Module enhances feature maps and increases classification accuracy.

Algorithms

The five-stage garbage classification architecture based on RepVGG is depicted in Fig. 5. Convolutional layers are used to extract features after input images. After stage 3, a FEMA module improves features, and prior to stage 5, NGCAM pays attention. Lastly, the waste category is produced by a fully connected layer and global average pooling.

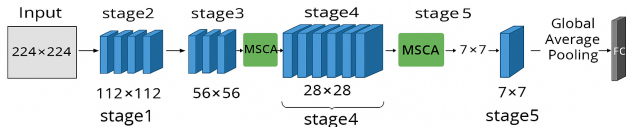


Fig. 5: Architecture of RepVGG with Multi-Scale Channel Attention (MSCA)

RepVGG-MEM (RepVGG + MSCA):

The paper’s main algorithm, RepVGG-MEM, is a lightweight convolutional neural network (CNN) for garbage classification. It is based on the RepVGG architecture, which is well-known for its structural reparameterization, which efficiently combines multi-branch topologies (such as 1×1 and 3×3 convolutions) into a single 3×3 convolution during inference. By emphasizing multi-scale channel relationships, the MEM

enhancement incorporates the Multi-Scale Channel Attention (MSCA) module to enhance feature extraction while striking a balance between speed (0.55G FLOPs in the pruned RepVGG-MEM5) and accuracy (93.26% unpruned, 92.09% pruned). For low-end devices, the model outperforms ResNet50, MobileNetV3, VGG16, and Xception in terms of efficiency. The goal is to offer a garbage classification model that is quick, accurate, and small enough to fit into environments with limited resources.

Basic Steps in an Algorithm:

Preprocessing of Input: Apply augmentations to the input image (224×224×3), such as Mixup.

Multi-Branch Convolution: Use an identity branch in addition to batch normalization and ReLU to process the input through parallel 1×1 and 3×3 convolutional layers during training.

Reparameterization: At inference, remove branch overhead by summing the weights and combining the branches into a single 3×3 convolution.

Application for MSCA: Scale the feature maps after calculating channel-wise attention weights with sigmoid activation, bottleneck (e.g., 1/16 channels), and global average pooling.

Pooling and Classification: To reduce spatial dimensions, use global average pooling. Then, for 6-class classification (e.g., plastic, paper, etc.), a fully connected layer with softmax is applied.

A. Exponential Moving Average (EMA)

EMA is a training technique that improves stability and generalization by maintaining a moving average of model parameters. It is used as part of the parameter smoothing procedure (discussed on Page 7) in the paper, most likely during fine-tuning following pruning. This contributes to the pruned RepVGG-MEM5’s 92.09% accuracy by reducing the accuracy drop brought on by aggressive pruning. The goal is to improve model robustness, particularly after pruning, and streamline parameter updates.

a) Basic Steps in an Algorithm::

- 1) Choose a smoothing rate (decay factor), usually near 1 (0.999).
- 2) Decide which neural network model you want to monitor during training.
- 3) To store the moving averages of each trainable parameter, create a dictionary named the “shadow model.”
- 4) Build a second dictionary to hold the model’s active weights for recovery at a later time.
- 5) Examine every trainable layer in the model and record its initial configuration in the shadow dictionary.
- 6) Repeat these steps during each weight update (usually after backpropagation):
- 7) For each training parameter:
- 8) To update the shadow weight, use the formula below:

$$\text{new_shadow} = (1 - \text{smoothing_rate}) \times \text{current_weight} + \text{smoothing_rate} \times \text{previous_shadow}$$

- 9) When you want to test or confirm your model's behavior:
- 10) Maintain the most recent model weights in the backup dictionary.
- 11) Enter the correct value for each model parameter in the shadow dictionary.
- 12) Following the completion of validation:
- 13) Use the backup to restore the model's original weights.
- 14) Delete the contents of the backup to reset the memory.
- 15) The updated shadow values now reflect the smoothed version of your model, ready for deployment or final evaluation.

B. Model Pruning: Magnitude-Based Pruning

Eliminates convolutional layer filters with the smallest L1 norm. Pruning is done in several rounds using an iterative approach.

a) Basic Steps in an Algorithm::

- 1) Initialize pruning iteration counter: $t \leftarrow 1$
- 2) Set pruning ratio $r \leftarrow a\%$
- 3) Define the maximum number of pruning iterations: k (usually $k = 1$ for one-shot pruning)
- 4) Repeat while $t \leq k$:
 - a) For each target convolutional layer in selected stages (stage3, stage4):
 - i) Compute L1-norm of each filter: $L1_norm[i] = \sum |W_i|$ over all elements of filter i
 - ii) Sort filters in ascending order based on their L1-norm values
 - iii) Select the bottom $a\%$ filters as candidates for pruning:
 $R = \{W_i \mid i \in \text{lowest } a\% \text{ of } L1_norm\}$
 - iv) Prune selected filters by zeroing out W_i and their corresponding biases (if applicable)
 - b) Evaluate model performance after pruning (e.g., accuracy, loss)
 - c) Fine-tune the pruned model for several training epochs (15 epochs) to recover lost accuracy
 - d) Re-evaluate the fine-tuned model on the validation/test set (accuracy, F1-score)
- 5) Increment pruning iteration counter: $t \leftarrow t + 1$
- 6) End While loop
- 7) Return the final fine-tuned and pruned model weights.

IV. EXPERIMENTAL SETUP

Using publicly accessible waste image datasets, we conducted a number of experiments to assess the suggested RepVGG-MEM model's performance. These datasets cover a wide range of common waste categories, including organic materials, paper, cardboard, plastic, metal, and glass. To guarantee network input compatibility, each image was resized to a consistent 224×224 pixel resolution. To ensure a balanced distribution across all classes, the dataset was randomly split into three subsets: 60% for training, 20% for validation, and 20% for testing.

To speed up the computational processes, the training was conducted on a system that had an NVIDIA GPU installed.

With an initial learning rate of 0.001, we employed the Adam optimizer. During training, methods such as AutoAugment policies and Mixup data augmentation were used to improve the model's capacity for generalization. Additionally, to stabilize learning and increase the accuracy of the final evaluation, an Exponential Moving Average (EMA) of the model weights was maintained. In order to speed up inference and decrease model size without compromising classification performance, structured pruning was employed.

V. RESULTS AND DISCUSSIONS

Using common performance metrics like accuracy, F1-score, and confusion matrix analysis, the suggested RepVGG-MEM model's performance was thoroughly assessed on a 14-class garbage classification dataset.

A. Confusion Matrix Analysis

The proposed RepVGG-MEM model's classification ability across 14 waste categories is shown in detail in the confusion matrix shown in Figure 6. With 4,129 correct classifications, the clothing category, for example, exhibits an exceptionally high true positive count, demonstrating the model's capacity to successfully learn unique characteristics of that class. The categories of paper, shoes, and white glass also show high accuracy, indicating consistent performance for both recyclable and non-recyclable materials. However, there are some misclassifications, mostly between classes that are visually similar.

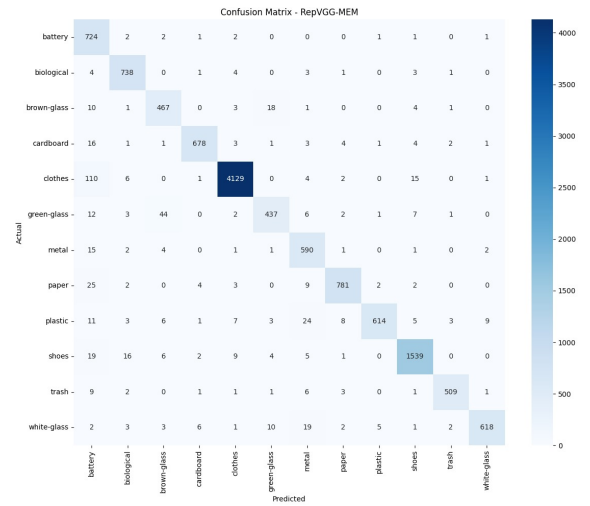


Fig. 6: Confusion matrix showing class-wise performance of the RepVGG-MEM model.

B. Accuracy Comparison Graph Analysis

Different RepVGG-based model variants' test accuracy performance is graphically depicted in the accuracy comparison graph (Figure 7). As can be seen, the baseline RepVGG model performs moderately, but adding strategies like Mixup and EMA (Exponential Moving Average) results in small gains. Nevertheless, the RepVGG-MEM architecture represents a significant advancement, surpassing all previous iterations.

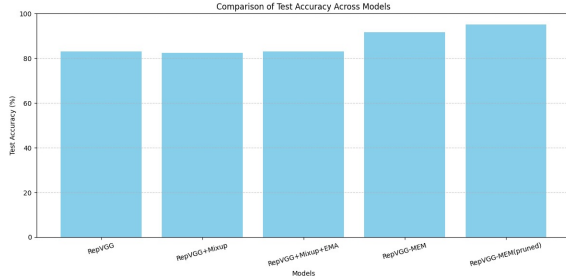


Fig. 7: Bar chart comparing test accuracy of RepVGG variants with and without enhancements.

Table II shows that the baseline RepVGG models achieved about 83% in accuracy and F1-score. Incorporating EMA, MSCA and Mixup (RepVGG-MEM) improved results to over 91%, while pruning delivered the best performance with 95.06% accuracy and 94.5% F1-score, along with strong secondary outcomes.

TABLE II: Accuracy and F1-Score of RepVGG Variants

Model	Accuracy (%)	F1-score (%)
RepVGG	83.4	82.7
RepVGG-M	83.2	82.9
RepVGG-ME	83.7	83.4
RepVGG-MEM	91.77	92.9
RepVGG-MEM (pruned)	95.06,[93.26]	94.5, [92.47]

VI. CONCLUSION

This work introduced a portable and effective RepVGG-based framework for real-time garbage classification. The main novelty lies in the integration of Exponential Moving Average (EMA), Mixup augmentation, and structured pruning within the RepVGG backbone, forming a single compact model. Unlike earlier studies that applied these strategies independently, our unified framework improves stability during training, enhances generalization across diverse waste categories, and reduces computational cost without compromising accuracy. Experiments on a combined dataset derived from TrashNet and TrashBox demonstrated a classification accuracy of 95.06%, which is 1.74% higher than the baseline RepVGG-MEM.

VII. LIMITATIONS

This study has a few limitations. The dataset is imbalanced, which lowers accuracy for rare waste types, and its reliance on open-source images may not fully represent real conditions such as poor lighting or mixed waste. The model covers only fixed categories, requiring retraining for new classes, and despite being lightweight, training still needs significant computational effort. In addition, issues like dust and overlapping items were not considered, which could affect real-world performance.

VIII. FUTURE WORK

Future work will focus on improving model performance and applicability. This includes expanding the dataset to cover

more waste categories and diverse real-world conditions, enhancing the model to handle overlapping or damaged items, and optimizing it for deployment on low-powered devices. Additionally, integrating real-time waste monitoring and exploring hybrid approaches that combine visual recognition with sensor data could further improve accuracy and practical utility.

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